

COURSE: EEE 4484 (DIGITAL ELECTRONICS & PULSE TECHNIQUES LAB.)
EXPERIMENT NO. 01
STUDY OF DIFFERENT LOGIC FAMILY

OBJECTIVE:

1. To analyze the circuits of different logic families specially DL (Diode Logic) and DTL (Diode-Transistor Logic) gates.
2. Study of a transistorized NOT gate.
3. To create a gate-level implementation of a combinational logic function described by a truth table using AND/OR/NOT gates.

INTRODUCTION:

A logic family is a group of electronic logic gates constructed using one of several different designs, usually with compatible logic levels and power supply characteristics within a family. The basic building blocks of logic family circuits are logic gates. And logic gates themselves are simple electronic circuits comprising of diodes, transistors and resistors. Thus, logic families can be categorized into different parts as follows:

1. Diode logic (DL),
2. Diode Transistor Logic (DTL),
3. Resistor Transistor Logic (RTL),
4. Transistor-Transistor Logic (TTL),
5. Integrated Injection Logic (IIL), and many more.

EQUIPMENT:

1. DC Power Supply,
2. Breadboard,
3. Resistors,
4. Diodes,
5. Transistors (BJTs),
6. Multimeter, and
7. Connecting wires.

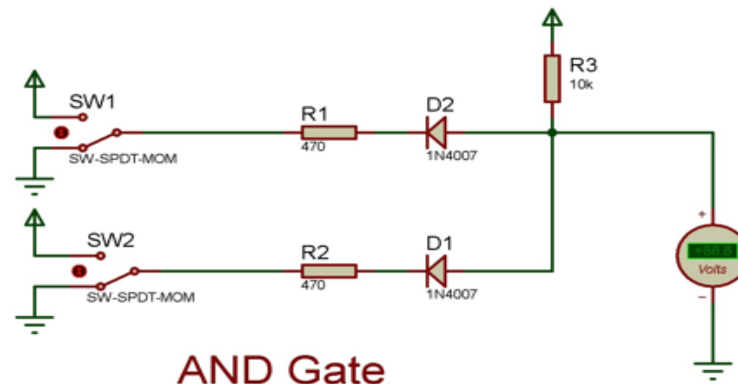
KNOWLEDGE PROFILE AND APPLICATION:

1. Operation of Basic Logic Gates (AND, OR, NAND, NOR, NOT),
2. Biasing of Diode, and
3. Common Emitter Configuration of BJT.

PROCEDURE:

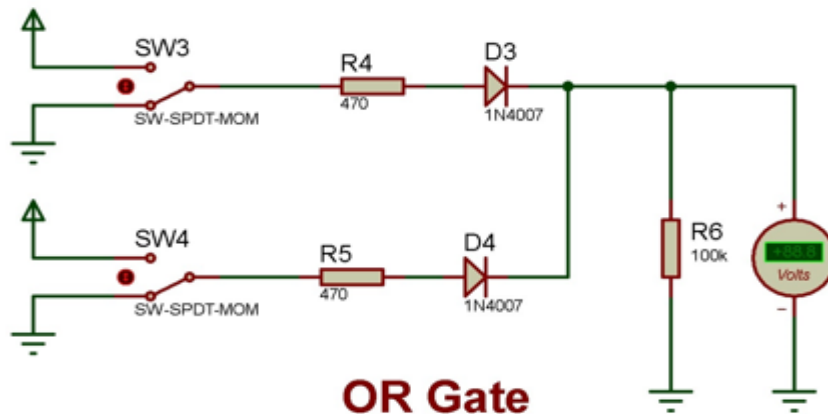
1. Connect the following circuits given in Fig. 1(a), Fig. 1(b), Fig. 1(c), Fig. 1(d) and Fig. 1(e) in breadboard.
2. Measure the output voltage using multimeter and fill up the corresponding truth tables.
3. Use +5V power supply from the board to connect the switches.

CIRCUIT DIAGRAM & TRUTH TABLE:



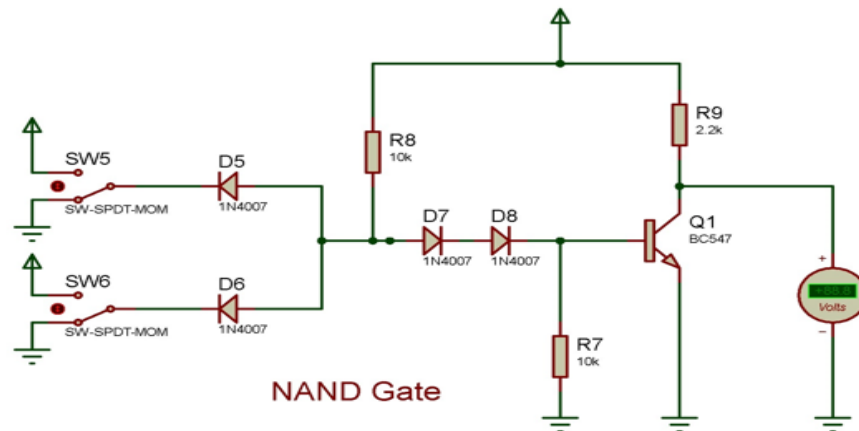
Input 1	Input 2	Output Voltage (Vo)	Logic Output Level
0	0		
0	1		
1	0		
1	1		

Fig. 1(a): AND Gate



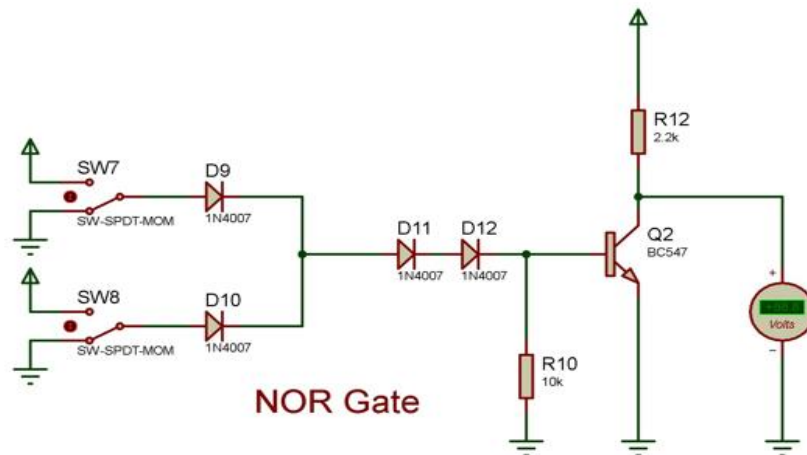
Input 1	Input 2	Output Voltage (Vo)	Logic Output Level
0	0		
0	1		
1	0		
1	1		

Fig. 1(b): OR Gate



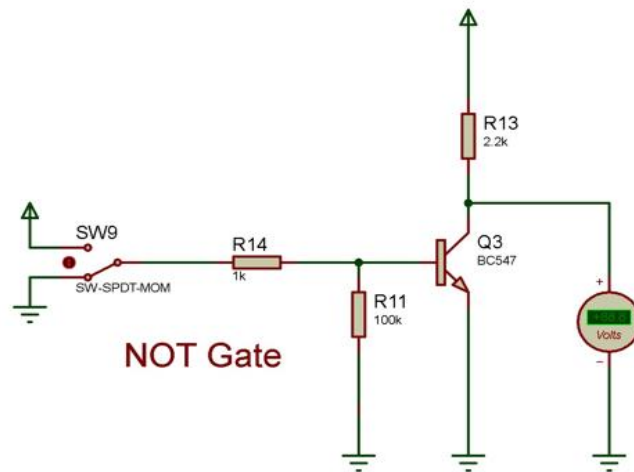
Input 1	Input 2	Output Voltage (Vo)	Logic Output Level
0	0		
0	1		
1	0		
1	1		

Fig. 1(c): NAND Gate



Input 1	Input 2	Output Voltage (Vo)	Logic Output Level
0	0		
0	1		
1	0		
1	1		

Fig. 1(d): NOR Gate



Input	Output Voltage (Vo)	Logic Output Level
0		
1		

Fig. 1(e): NOT Gate

REPORT QUESTION:

1. Analyze all the circuits with the help of truth table for positive logic.
2. Why two diodes have been used in the circuit connection of Fig. 1(d)? Explain.
3. What is the minimum voltage required to make the diodes forward biased in Fig. 1(a), Fig. 1(b), Fig. 1(c) and Fig. 1(d)?
4. Describe the operation of the circuit diagram in Fig. 1(e).